

Lumio

Clinical Intelligence for Mental Health

Business Context:

Problem Statement & Solution Architecture

Google Calendar & Google Meet Scheduling Integration

Core Problems to Solve:

1. PROBLEM STATEMENT

1. Users need one click login without maintaining a separate app password.
2. Every meeting needs an auto-generated video link (no manual Meet/Zoom link creation).
3. Multiple clients/guests must be invited and notified by email automatically.
4. Meeting must be visible both in the user's personal Google Calendar AND inside the app.

Functional Requirements

#	
Requirement	Priority
FR-1 User authentication via Google OAuth 2.0 (no separate username/password store)	Critical
FR-2 Auto-generate a unique Google Meet link for every meeting created	Critical
FR-3 Support inviting multiple client/guest emails per meeting	High
FR-4 Persist every meeting in Lumio's own custom calendar (source of truth)	Critical
FR-5 Sync meeting + Meet link into organizer's personal Google Calendar	High
FR-6 Schedule and display all times in Indian Standard Time (IST, GMT+5:30)	Medium

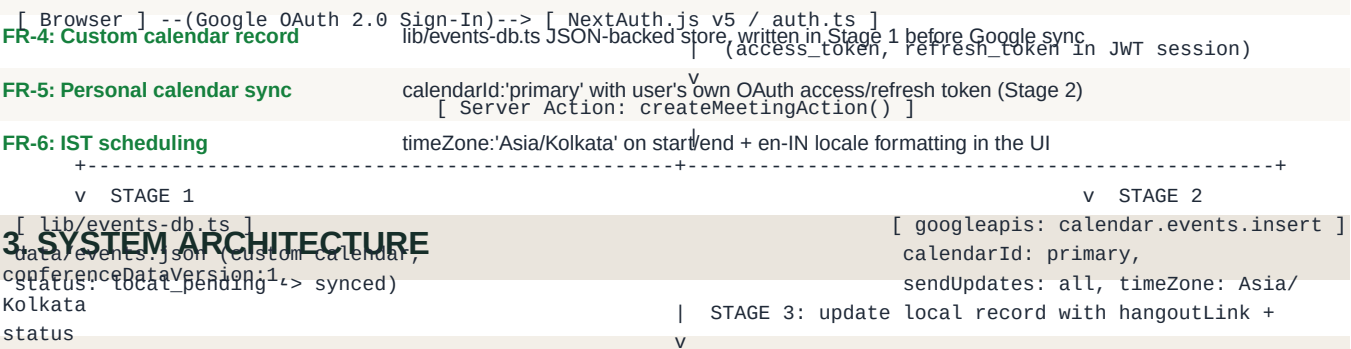
Approach:

Built a Next.js 15 application using Google OAuth 2.0 as the sole sign-in method (no local password store). Each signed-in user's Google refresh token is used server-side to call the Google Calendar API directly on their behalf, so events are created under their own identity as Organizer. Every meeting is written to Lumio's local custom-calendar database first, then synced to Google Calendar with conferenceDataVersion:1 to auto-provision a Google Meet link, and sendUpdates:'all' to email every guest.

Requirement !' Solution Mapping

Requirement	Implementation
FR-1: Google OAuth login	NextAuth.js v5 Google Provider, access_type=offline, prompt=consent (auth.ts)
FR-2: Auto Meet link	conferenceDataVersion:1 + conferenceSolutionKey hangoutsMeet on events.insert

COMPONENT & DATA-FLOW DIAGRAM



3. SYSTEM ARCHITECTURE

Result: Google generates the Meet link, emails all invited guests directly from the organizer's own Gmail account, and Lumio's UI (components/CustomCalendarView.tsx) renders the same meeting from the local database with a live status badge and Join Meet button.

Technology Stack

Framework	Next.js 15 (App Router, Server Actions)
Auth	NextAuth.js v5 - Google OAuth 2.0 (offline + consent)
Calendar API	googleapis v144 - Google Calendar API v3
Local Store	JSON file store (data/events.json)
UI	React + Tailwind CSS v4, Lucide icons

- Creating a meeting instantly appears in the Custom In-App Calendar (Grid + Timeline views).
- Google auto-generates a working <https://meet.google.com> link on every event.

- All added guest emails receive a native Google Calendar invite with the Meet link attached.

4. OUTCOME & VERIFICATION

- Meeting appears on the organizer's personal Google Calendar, correctly timed in IST.

- Verified live on Vercel production deployment with real Google account sign-in.

Live References

Repository: <https://github.com/kishorsumathi/lumio-calendar-poc>

Production: <https://google-calander-hv6k04gxy-vanco.vercel.app>

